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Another gem in matrix theory, <u>not</u> widely known.

Of course, $AB = BA \Rightarrow \sigma(AB) = \sigma(BA)$
What if $AB \neq BA$? Can we compare spectra?

<u>Theorem</u> Let $A \in M_{m,n}$, $B \in M_{n,m}$, $m \leq n$.
Then $AB \in M_m$, $BA \in M_n$ have the same eigenvalues counting multiplicities, together with BA having an extra $n-m$ zero eigenvalues. That is,

$$p_{BA}(t) = t^{n-m} p_{AB}(t)$$

In particular, if $m = n$ and at least one of A or B is invertible, then AB & BA are in fact similar!

Proof $\longrightarrow$

<u>Proof</u> ("Tricky", by Ludwig Elsner)

Observe the following matrix identity:

$$\underbrace{\begin{bmatrix} AB & 0 \\ B & 0 \end{bmatrix}}_{(m+n)\times(m+n)} \begin{bmatrix} I_m & A \\ 0 & I_n \end{bmatrix} = \begin{bmatrix} AB & ABA \\ B & BA \end{bmatrix}$$

$$\shortparallel$$

$$\begin{bmatrix} I_m & A \\ 0 & I_n \end{bmatrix} \begin{bmatrix} 0 & 0 \\ B & BA \end{bmatrix} = \begin{bmatrix} AB & ABA \\ B & BA \end{bmatrix}$$

So

$$\boxed{\begin{bmatrix} I_m & A \\ 0 & I_n \end{bmatrix}^{-1} \begin{bmatrix} AB & 0 \\ B & 0 \end{bmatrix} \begin{bmatrix} I_m & A \\ 0 & I_n \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ B & BA \end{bmatrix}}$$

inverse exists!

i.e. $\overset{X}{\begin{bmatrix} AB & 0 \\ B & 0 \end{bmatrix}}$ & $\overset{Y}{\begin{bmatrix} 0 & 0 \\ B & BA \end{bmatrix}}$

are similar!

So $\sigma(AB) \subseteq \sigma(X)$

$\sigma(X) = \sigma(AB) \cup \{\underbrace{0, \dots, 0}_{n \text{ of them}}\}$

$\sigma(BA) \subseteq \sigma(Y)$

$\sigma(Y) = \sigma(BA) \cup \{\underbrace{0, \dots, 0}_{m \text{ of them}}\}$

$\sigma(X) = \sigma(Y)$

So $\sigma(BA) = \sigma(AB) \cup \{\underbrace{0, \dots, 0}_{n-m \text{ of them}}\}$

If $m = n$ & say A^{-1} exists, then

$$A^{-1}(AB)A = BA$$

$\downarrow$

similarity of AB

so $BA \sim AB$ ■

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§ 1.4 Eigenvectors

<u>Observations</u> Let $A \in M_n$, then

(i) $\sigma(A) = \sigma(A^T)$, (ii) $\sigma(A^*) = \overline{\sigma(A)}$

$Ax = \lambda x$, $\lambda \in \sigma(A)$, $x \neq 0$

$\iff \det(A - \lambda I) = 0$

$\iff \det((A - \lambda I)^T) = 0$

$\iff \det(A^T - \lambda I) = 0$

$\iff \lambda \in \sigma(A^T)$

$\lambda \in \sigma(A)$

$\iff \det(A - \lambda I) = 0$

$\iff \det((A - \lambda I)^*) = 0$

$\iff \det(A^* - \bar{\lambda} I) = 0$

$\iff \bar{\lambda} \in \sigma(A^*)$

Definitions

Given $A \in M_n$, $\lambda \in \sigma(A)$, $N(A-\lambda I)$ (or $\mathrm{Null}(A-\lambda I)$) is called the <u>eigenspace</u> of A corresponding to λ, which is a subspace of $\mathbb{C}^n$. We call

$$\dim N(A-\lambda I)$$

the <u>geometric multiplicity</u> of $\lambda \in \sigma(A)$.

The multiplicity of λ as a root of the characteristic polynomial, is called the <u>algebraic multiplicity</u> of $\lambda \in \sigma(A)$.

Denote (informally)

$$am(\lambda) = \text{alg mult}$$
$$gm(\lambda) = \text{geom mult.}$$

Example $\longrightarrow$

<u>Example</u> $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $\sigma(A) = \{0, 0\}$

So $am(0) = 2$

To find $gm(0)$, solve

$$(A - 0I)x = 0$$

(from $N(A - 0I) = N(A)$)

basis

$$\left[\begin{array}{cc|c} 0 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right] \qquad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad x_2 = 0, \quad x_1 = \text{free}$$

i.e.,

$$N(A - 0I) = \text{span}\left\{\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right\}$$

So $gm(0) = 1 < 2 = am(0)$

<u>Theorem</u> Let $A \in M_n$, $\lambda \in \sigma(A)$. Then

$$gm(\lambda) \leq am(\lambda)$$

<u>Proof</u> Say $\underbrace{\dim N(A - \lambda I)}_{gm(\lambda)} = k \leq n$. Then

there exists basis for $N(A - \lambda I)$:

$$\{x^{(1)}, x^{(2)}, \ldots, x^{(k)}\}. \leftarrow \text{eigenvectors} \sim \lambda$$

Extend this to a basis for $\mathbb{C}^n$:

$$\{x^{(1)}, \dots, x^{(k)}, y^{(1)}, \dots, y^{(n-k)}\}$$

Form

$$S = [x^{(1)} | x^{(2)} | \dots | x^{(k)} | y^{(1)} | \dots | y^{(n-k)}]$$

$\in M_n$ invertible.

Then

$$AS = [Ax^{(1)} | Ax^{(2)} | \dots | Ax^{(k)} | Ay^{(1)} | \dots | Ay^{(n-k)}]$$

$$= [\lambda x^{(1)} | \lambda x^{(2)} | \dots | \lambda x^{(k)} | Ay^{(1)} | \dots | Ay^{(n-k)}]$$

$$\overset{!}{=} S \begin{bmatrix} \overbrace{\boxed{\begin{matrix} \lambda & & 0 \\ & \ddots & \\ 0 & & \lambda \end{matrix}}}^{\lambda I_k} & * \\ 0 & \underbrace{*}_{\text{something}} \end{bmatrix}$$

$$\Rightarrow S^{-1}AS = \begin{bmatrix} \boxed{\lambda I_k} & * \\ 0 & \boxed{*} \end{bmatrix}$$

So $\lambda \in \sigma(S^{-1}AS) = \sigma(A)$ repeated at least k times, i.e.

$$\mathrm{am}(\lambda) \geq k = \mathrm{gm}(\lambda)$$

∎

Late facts (reminders)

> $A = \begin{bmatrix} d_1 & & & 0 \\ & d_2 & & \\ & & \ddots & \\ 0 & & & d_n \end{bmatrix} \in M_n$, $A = \mathrm{diag}(d_1, d_2, \dots, d_n)$

diagonal matrix

> $A = \begin{bmatrix} d_1 & & & * \\ & d_2 & & \\ & & \ddots & \\ 0 & & & d_n \end{bmatrix} \in M_n$, upper triangular

> $A = \begin{bmatrix} d_1 & & & 0 \\ & d_2 & & \\ & & \ddots & \\ * & & & d_n \end{bmatrix} \in M_n$, lower triangular

In all cases, $\sigma(A) = \{d_1, d_2, \dots, d_n\}$

(prove!)

> **Block-triangular matrix** (also block-diagonal)

$$A = \begin{bmatrix} \boxed{A_{11}} & & & * \\ & \boxed{A_{22}} & & \\ & & \ddots & \\ 0 & & & \boxed{A_{kk}} \end{bmatrix} \in M_n, \quad A_{jj} \in M_{n_j}, \quad n_1 + \dots + n_k = n$$

(not necessarily triangular)

eg.

$$\begin{bmatrix} \boxed{\begin{matrix} 1 & 4 \\ 1 & 2 \end{matrix}} & & & * \\ & \boxed{\begin{matrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{matrix}} & & \\ & & \boxed{1} & \\ 0 & & & \boxed{\begin{matrix} 1 & 2 \\ 1 & 3 \end{matrix}} \end{bmatrix} \in M_8$$

<u>Facts</u>: $\sigma(A) = \sigma(A_{11}) \cup \sigma(A_{22}) \cup \cdots \cup \sigma(A_{kk})$

$\det A = \det A_{11} \det A_{22} \cdots \det A_{kk}$

(prove in at least two different ways)

Terminology

- If $\forall \lambda \in \sigma(A)$, $gm(\lambda) = 1$, we call A <u>nondegatory</u>.

- We call A a <u>defective</u> matrix if $\exists \lambda \in \sigma(A)$ such that $am(\lambda) > gm(\lambda)$. We call this λ a <u>defective eigenvalue</u>.

<u>So</u>

- nondegatory & nondefective
  $\iff \forall \lambda \in \sigma(A) : am(\lambda) = gm(\lambda) = 1$
  $\iff$ distinct eigenvalues
  $\Rightarrow$ diagonalizable

- non-defective $\iff am(\lambda) = gm(\lambda), \forall \lambda \in \sigma(A)$
  $\iff$ diagonalizable. $\xrightarrow{\text{Proof}}$

<u>Proof</u> $A \in M_n$

$$n = \sum_{\lambda \in \sigma(A)} am(\lambda) \overset{am(\lambda)=gm(\lambda)}{=\!=\!=} \sum_{\lambda \in \sigma(A)} gm(\lambda)$$

Thus A has n lin. indep. eigenvectors, i.e. by theorem, A is diagonalizable.

LEFT EIGENVECTORS / EIGENSPACES

<u>Definition</u> Vector $y \in \mathbb{C}^n$, $y \neq 0$, is called a <u>left-eigenvector</u> of $A \in M_n$ corresponding to $\lambda \in \sigma(A)$ if

$$y^* A = \lambda y^*$$

$$(\lambda \in \sigma(A) \Rightarrow \exists x \neq 0 : Ax = \lambda x)$$

Existential problem

(why do left-eigenvectors always exist?)

$\lambda \in \sigma(A) \implies \bar{\lambda} \in \sigma(A^*)$ (because $Ax = \lambda x \implies x^* A^* = \bar{\lambda} x^* \implies x^*(A^* - \bar{\lambda} I) = 0 \implies A^* - \bar{\lambda} I$ is singular $\implies \bar{\lambda} \in \sigma(A^*)$)

$\implies \exists\, y \neq 0 : A^* y = \bar{\lambda} y$

$\implies (A^* y)^* = (\bar{\lambda} y)^*$

$\implies y^* A = \bar{\bar{\lambda}} y^* = \lambda y^*$ ✓

"Knowing" a matrix, means knowing the eigenspaces, & the left-eigenspaces

$N(A - \lambda I)$ $\qquad$ $N(A^* - \lambda I)$

$\forall \lambda \in \sigma(A)$

Theorem If $A \in M_n$, $\lambda, \mu \in \sigma(A)$, $\lambda \neq \mu$.

and if x is an eigenvector of A corresp. to λ

y ——— left eigenvector —"— ——— μ

then

$$y^* x = 0 \qquad (\text{or } x^* y = 0)$$

(i.e. x, y are "orthogonal")

Proof

Assumptions are: $\begin{cases} Ax = \lambda x, \ x \neq 0 \\ y^* A = \mu y^*, \ y \neq 0 \\ \lambda \neq \mu \end{cases}$

Then

$$y^* A x \begin{cases} y^* \lambda x = \lambda y^* x \\ \mu y^* x \end{cases}$$

So $\lambda y^* x = \mu y^* x$

i.e. $(\lambda - \mu) x^* y = 0 \qquad \lambda - \mu \neq 0$

i.e. $x^* y = 0 \qquad (y^* x = 0)$ ∎

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CHAPTER 2 : UNITARY EQUIVALENCE & NORMAL MATRICES

Terminology & some quick refresher on inner products

- $A \in M_n$, $S \in M_n$ nonsingular. The transformation

$$A \longrightarrow S^* A S$$

is <u>$*$-congruence</u>. $\quad (S^* = (\overline{S^T}))$

- Recall (review). Given $x, y \in \mathbb{C}^n$, the quantity

$$\langle x, y \rangle = x^* y$$

is an "inner product" in $\mathbb{C}^n$ that satisfies

- $x^* y \in \mathbb{C}$
- $x^* x \geq 0$, $x^* x > 0 \iff x \neq 0$
- length of $x = \|x\|_2 =$ Euclidean length $= \sqrt{x^* x} =$ 2-norm

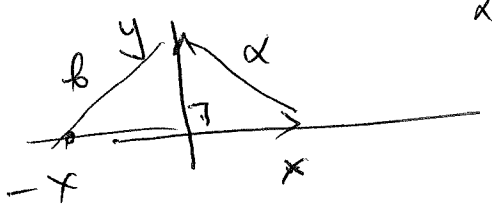
<u>Example</u> $x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{C}^n$

$$\|x\|_2 = \sqrt{x^* x} = \left([\bar{x}_1 \ \bar{x}_2 \cdots \bar{x}_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \right)^{1/2}$$

$$= \left(\sum_{j=1}^{n} x_j \bar{x}_j \right)^{1/2}$$

<u>Defn</u>: We say $x, y \in \mathbb{C}^n$ are <u>orthogonal</u> if

$$x^* y = 0$$



$\alpha = b \longleftrightarrow x^* y = 0$

$\alpha = \|x+y\|_2$

$b = \|y-x\|_2$

<u>Recall</u>

$$\|x+y\|_2 \le \|x\|_2 + \|y\|_2 \quad \text{(triangle)}$$

$$|x^* y| \le \|x\|_2 \|y\|_2 \quad \text{(Cauchy-Schwartz)}$$

<u>Defn</u> A set $\{v_1, v_2, \ldots, v_k\} \subseteq \mathbb{C}^n$ is called <u>orthogonal</u> if $v_i^* v_j = 0$ for all $i \neq j$, $i, j \in \{1, 2, \ldots, k\}$

The set is <u>orthonormal</u> if it is orthogonal and $\|v_j\|_2 = 1$, $j = 1, 2, \ldots, n$.

Theorem (prove!)

An orthogonal set in $\mathbb{C}^n$ of nonzero vectors (e.g. an orthonormal set) is linearly independent.

§ 2.1 Unitary matrices

Definition Matrix $U \in M_n$ is called unitary if $U^* U = I$

(i.e. $U^* = U^{-1}$, so $UU^* = I$)

(in $M_n(\mathbb{R})$ unitary = orthogonal matrix $\downarrow U^T = U^{-1}$)

Inverse of unitary is easy to find, but lots more nice things are true:

<u>Theorem</u> Let $U \in M_n$. The following are equivalent:

(i) U is unitary ($U^*U = I$)

(ii) U is nonsingular & $U^{-1} = U^*$

(iii) $UU^* = I$

(iv) U^* is unitary

(v) Column of U form an orthonormal set

(vi) Rows of U ——— " ———

(vii) $\forall x \in \mathbb{C}^n$, $\|Ux\|_2 = \|x\|_2$

(multiplication by unitary preserves length)

<u>Proof</u> Equivalence (i) – (iv) is clearly based on just definitions.

Look at (v): ((v) <=> (i))

$$U = [u_1\ u_2 \dots u_n] \in M_n \text{ unitary}$$

columns

$$\left(U^* = \begin{bmatrix} \overline{u_1^T} \\ \overline{u_2^T} \\ \vdots \\ \overline{u_n^T} \end{bmatrix} \right)$$

<u>Compute</u>

$$U^* U = \begin{bmatrix} \overline{u_1}^T \\ \vdots \\ \overline{u_n}^T \end{bmatrix} [u_1 \cdots u_n]$$

$$= \begin{bmatrix} \overline{u_1}^T u_1 & \overline{u_1}^T u_2 & \cdots & \overline{u_1}^T u_n \\ \overline{u_2}^T u_1 & \overline{u_2}^T u_2 & \cdots & \overline{u_2}^T u_n \\ \cdots & \cdots & \cdots & \cdots \\ \overline{u_n}^T u_1 & \overline{u_n}^T u_2 & \cdots & \overline{u_n}^T u_n \end{bmatrix}$$

$$\overset{\text{know}}{=} \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & 0 \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 \; 1 \end{bmatrix}$$

i.e. $\overline{u_j}^T u_j = 1$, $j = 1, 2, \ldots, n$

$\overline{u_j}^T u_i = 0$, $i \neq j$

i.e. $\{u_1, u_2, \ldots, u_n\}$ is orthonormal.

Converse is just reversal of arguments, so (i) $\Leftrightarrow$ (v).

Similarly, (i) $\Leftrightarrow$ (vi).

Left to do: (i) $\Leftrightarrow$ (vii)

(i) $\Rightarrow$ (vii). Let $x \in \mathbb{C}^n$. Then

$$\|Ux\|_2^2 = (Ux)^*(Ux)$$
$$= x^* \underbrace{U^* U} x$$
$$= x^* I x = x^* x = \|x\|_2^2$$

i.e. $\|Ux\|_2 = \|x\|_2$.

(vii) $\Rightarrow$ (i).

<u>Need lemma</u> If $A \in M_n$ and $x^* A x = 0$ for all $x \in \mathbb{C}^n$, then $A = 0$.
(iff)

sure? of general interest

<u>Proof</u> (nontrivial) $\rightarrow$ try!

We'll apply lemma to $A = U^* U - I$

TBC...